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Decoupling geographical constraints from human mobility

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Driven by access to large volumes of movement data, the study of human mobility has grown rapidly over the past few decades. The field has shown that human mobility is scale-free, proposed models to generate scale-free moving distance distributions and explained how the scale-free distribution arises. It has not, however, explicitly addressed how mobility is structured by geographical constraints, such as how mobility relates to the outlines of landmasses, lakes and rivers and the placement of buildings, roadways and cities. On the basis of millions of moves, we show how separating the effect of geography from mobility choices reveals a power law spanning five orders of magnitude. To do so, we incorporate geography via the pair distribution function, which encapsulates the structure of locations on which mobility occurs. By showing how the spatial distribution of human settlements shapes human mobility, our approach bridges the gap between distance- and opportunity-based models of human mobility.

Human mobility is a fundamental aspect of societies, with constant flows of individuals and goods circulating through cities and landscapes. This flow of individuals moving from place to place underpins our social, economic and cultural lives[1]. Understanding mobility patterns therefore enables us to mitigate the spread of infectious diseases[2,3], reduce pollution[4] or organize urban planning[5]. Despite the fact that human movement is both shaped by and shaping our geography[6,7], the field of data-driven human mobility has not considered these two processes together[8].

Geography trivially constrains human mobility. Consider two simple examples: two houses cannot occupy the same physical space, thereby imposing a minimum distance for any movement between them. On a larger scale, residents of an isolated island with a 10-km radius cannot travel distances greater than 20 km: they would end up in the sea.

Previous work has shown that human mobility is highly structured, with power-law distributions of distance[9,10] and trip frequency[11], inflows between administrative boundaries[12,13], conservation laws[14] and scales corresponding to structures of the built environment[15]. Simultaneously, the structure of the built environment is well understood, from the hierarchical organization of human settlements[16] at the scales of countries[17–19] to street-level networks[20,21], as well as the fractal geometries of cities[22,23] and how they developed[6].

A key concern, however, is that the work cited above arises from two literatures that have evolved separately. The recent surge of data-driven work on human mobility has not explicitly addressed how the observed mobility data are structured by the constraints set by geography[8]. In that sense, large-scale data-driven human mobility research[24] and the community of geographers[6,18] remain distinct fields with complementary approaches.

Within data-driven human mobility studies, there are two main theories for human movement[24]. The first theory considers physical distance to deter mobility[25,26], a concept often expressed through gravity models[12]. The alternative viewpoint suggests that human movement does not depend on distance but rather on the quantity of opportunities nearby[27,28].

There have been scattered efforts to incorporate a notion of geography in data-driven human mobility, but this work has been limited by incomplete data—for example, relying on coarse-grained data such as data in grids[29], administrative units[30] or artificial geographical space[31], obscuring smaller scales and reducing geography to discrete density maps[32,33]. The use of neural embeddings also achieves a similar effect[34] and leads to the emergence of gravity laws[35].

The question of how to incorporate geography in the data-driven approaches to human mobility therefore still lacks a comprehensive and satisfying answer. Here we address this gap and present a

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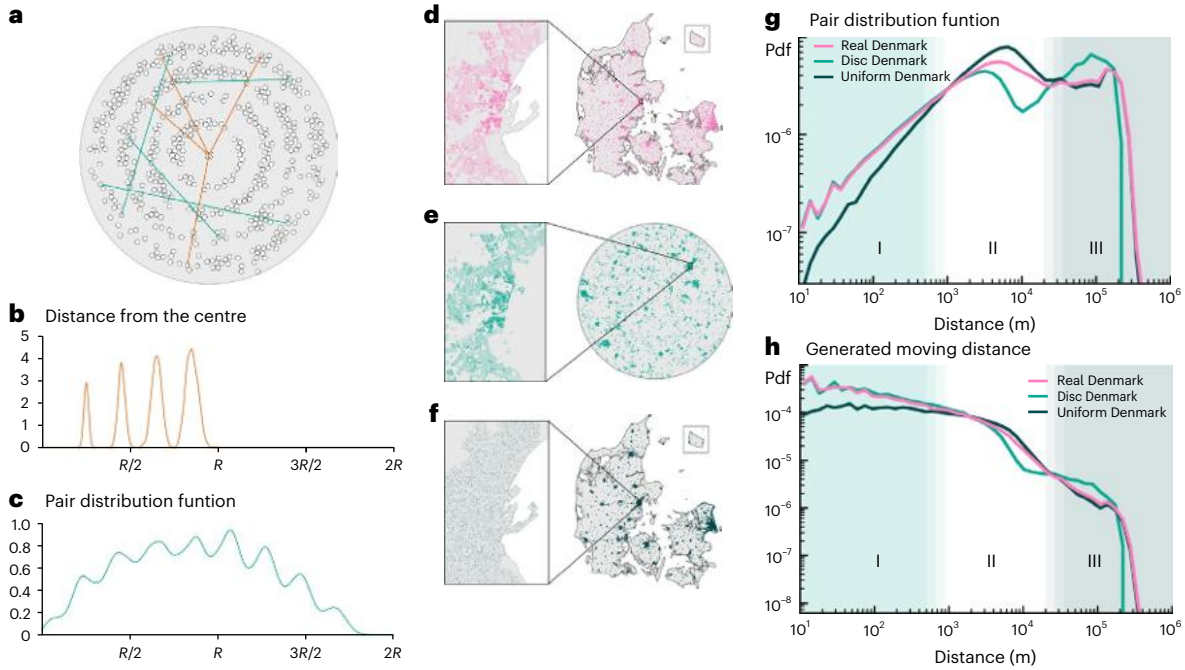


Fig. 1 | Illustration of geography's influence on human mobility. a, Spatial arrangement of concentric circles of locations. The white cross indicates the centre. **b**, Distance distribution of points to the centre of the disc (white cross in **a**; orange lines indicate example distances). **c**, Pair distribution of the points in the concentric circles (teal lines indicate example pairwise distances). Examples of more shapes and distributions of locations are studied in Supplementary Information section 1.1. **d**–**f**, Three geographies: Real Denmark, with 3,300,000 addresses (**d**); Disc Denmark, where the macro structure has cities distributed uniformly at random on a disc, but the microstructure of the cities is preserved (**e**); and Uniform Denmark, with the macro structure of Denmark (shape of landmasses and city positions) but no local microstructure (that is, uniformly distributed locations within cities) (**f**). **g**, Pair distribution (pdf) of the three geographies. **h**, The simulated observed moving distance for an intrinsic distance attractiveness of $1/r$. In **g** and **h**, i, ii and iii respectively represent the intra-city scale, the city scale and the inter-city scale.

framework to understand geography and the distribution of locations (addresses, points of interest and stop locations). We argue that the locations where people stop capture deep aspects of the mobility constraints set by geography: the built environment and urban structure, the placement of rivers and lakes, country outlines and so on.

Our primary analysis is based on a dataset comprising 36 years of registry data on 39,000,000 migrations between addresses in Denmark, pinpointed with an uncertainty of only 2 m and previously unused for studying human mobility. This resolution far exceeds the state-of-the-art 50-m accuracy typical of GPS data. Below, we extend the analysis to day-to-day mobility and points of interest.

Analysing these datasets using tools from statistical physics (for example, the pair distribution function) and by considering locations as particles, our approach unveils mobility patterns at scales that were previously obscured.

Results

The concept of the pair distribution function

To characterize geography, we studied the pair distribution function (hereafter 'pair distribution') between locations. The pair distribution is a powerful tool from statistical mechanics developed to understand the structural properties of materials[36]. Here we argue that the pair distribution is able to capture the structural properties of locations in a given geography. To get a sense of this distribution, consider a hypothetical spatial arrangement of concentric circles of n locations (stop locations, points of interest or addresses) on a disc of radius R (Fig. 1a). For an individual positioned at the centre of these circles (a white cross in the figure), the potential moves are constrained by the positions of other locations. For instance, in the absence of a location at distance $R/2$, a movement of that exact distance is impossible. To understand the potential movements from the disc's centre, we computed the distribution of distances of every location to the centre (Fig. 1b), which can be interpreted as what we would observe if an individual moved many times from the centre to one of the locations on the disc, choosing their new location independently of distance. We can place each of the n locations in the focus of such an analysis, finding a unique distance distribution for each of them, and calculate an average distance distribution. The result quantifies the expected number of locations found at distance r from an average location and is mathematically equivalent to the pair distribution, which simply encodes the number of location pairs found at distance r (Fig. 1c).

The observed data are a product of two separate components

This simple observation has a profound consequence. Any study of the observed movement distances $f(r)$ ends up analysing a biased set of observations: a quantity that includes the geometry of space Ω and the density $\varrho^d(x), x \in \Omega$, both reflected in the pair distribution $p(r)$ (see 'Pair distribution function' in Methods). We argue that any 'universal' behaviour of the system—that is, a geometry-independent law that encapsulates the behaviour of the system—must be captured entirely in a quantity that depends only on distance. We call this the intrinsic distance attractiveness.

Thus, to uncover any geometry-independent behaviour of our system, we must adjust our observation $f(r)$ by the geometry encoded in $p(r)$; that is:

$$\pi(r) \propto \frac{f(r)}{p(r)}. \quad (1)$$

To illustrate how geography shapes the pair distribution and the observed moving distances, we considered the mobility trace

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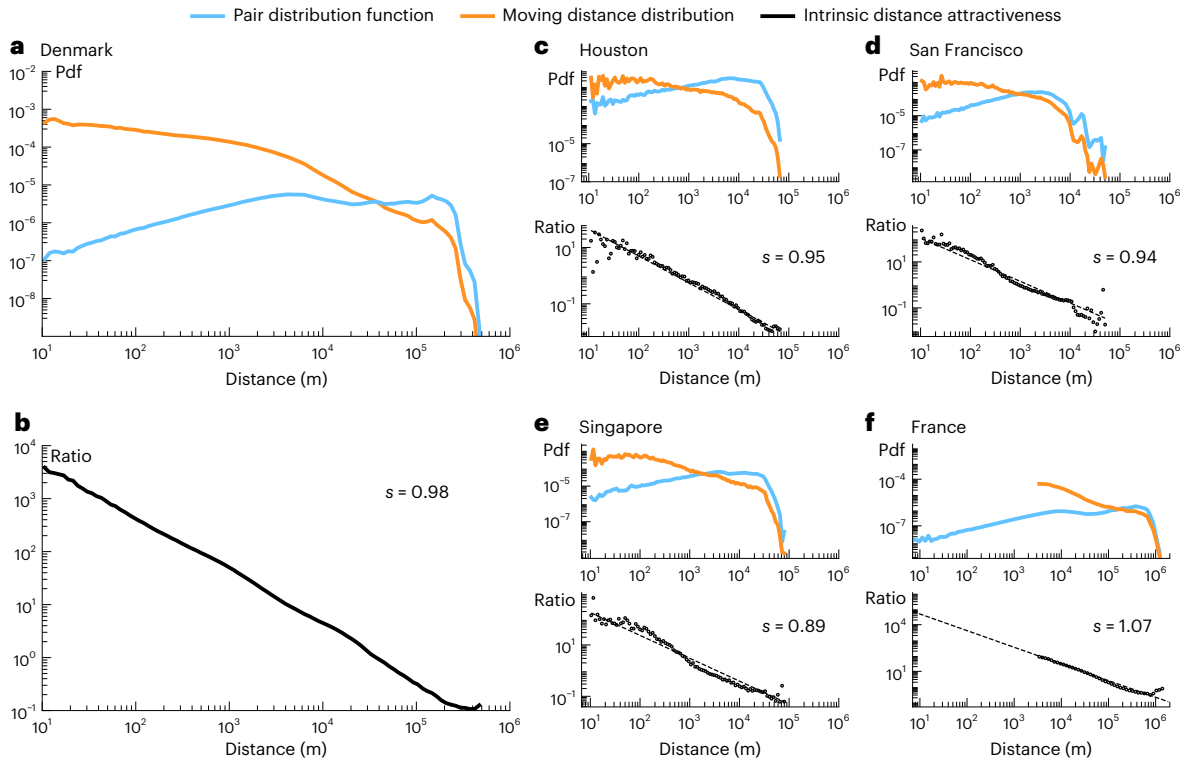


Fig. 2 | Renormalizing the moving distance distribution by the pair distribution uncovers a power law. a, Empirical moving distance distribution and pair distribution in Denmark. **b**, Dividing the mobility by the pair distribution yields an intrinsic distance attractiveness that follows a power law from 10 m to 500 km (maximum-likelihood estimation). **c**–**e**, Observed moving distance distribution and pair distribution for day-to-day mobility between points of interest in Houston (**c**), San Francisco (**d**) and Singapore (**e**). **f**, Observed moving distance distribution and pair distribution for residential mobility in France. The *y*-axis ranges vary across plots because the geographical areas under consideration differ in size.

simulation for three different geographies (Fig. 1d–f and Supplementary Information section 1.3). The first geography, 'Real Denmark', uses the actual geography of Denmark, including 3,300,000 precise locations (addresses) (Fig. 1d). The second, 'Disc Denmark', maintains the microstructure of cities but alters city positions and landmass shapes, distributing real city centres uniformly on a disc (Fig. 1e). The third, 'Uniform Denmark', keeps the macro structure of landmasses and city positions from Real Denmark but features uniformly dense cities (Fig. 1f).

The pair distribution effectively encapsulates the nuances of each geographical layout (Fig. 1g). On the local scales (less than 1 km), the pair distribution of the Real Denmark geography aligns with the one of Disc Denmark, reflecting city layouts. However, at broader scales (more than 10 km), the pair distribution of Real Denmark matches that of Uniform Denmark, indicating the influence of the shape of land masses and city positions.

To generate mobility traces, we imposed the same intrinsic distance attractiveness on each geography and made a gravity-law-inspired ansatz that it follows:

$$\pi(r) = 1/r. \tag{2}$$

Remarkably, despite having the same intrinsic distance attractiveness, the resulting observed mobility traces are distinct across the three Denmarks (Fig. 1h). This outcome is notable as it shows that even under a similar movement law, the variations in geography alone can lead to distinct mobility patterns.

A power law that spans five orders of magnitude

We next demonstrated the impact of taking the pair distribution into account, by first analysing an exhaustive dataset—39,000,000 residential moves between 3,300,000 Danish addresses over 36 years. The dataset has very low selection bias, as it covers all residential moves (see 'Data' in Methods). To uncover the intrinsic distance attractiveness of residential mobility, we first eliminated the geographical component by normalizing the observed moving distance distribution by the pair distribution (Fig. 2a). This normalization revealed that the intrinsic distance attractiveness function (equation (1)) follows a power law remarkably well:

$$\pi(r) = \frac{f(r)}{p(r)} = \frac{1}{r^s}. \tag{3}$$

This power law is consistent across scales ranging from 10 m to 500 km, covering five orders of magnitude (Fig. 2b). A maximum-likelihood fit provides an exponent of $s = 0.98$, which validates the ansatz in equation (2); the intrinsic distance attractiveness follows $\pi(r) = 1/r$. We consider this power law to be an extension of the gravity model to the continuous domain (see 'A "continuous" gravity law' in Methods)[37].

We found that the results above are particular neither to the geography of Denmark nor to residential mobility. Investigating several empirical datasets, Fig. 2c–e illustrates that the normalization by the pair distribution unveils power laws for day-to-day mobility across the diverse geographies of Houston, Singapore and San Francisco. We also found the same patterns for residential mobility in France (Fig. 2f). The empirical fact that our results take the same shape across countries and forms of mobility highlights the generality of the intrinsic distance attractiveness's functional shape: a power law that decreases with distance.

Next, we focus on the geographic regularities manifested through the power laws and the fluctuations in the pair distribution of Denmark (Fig. 2a).

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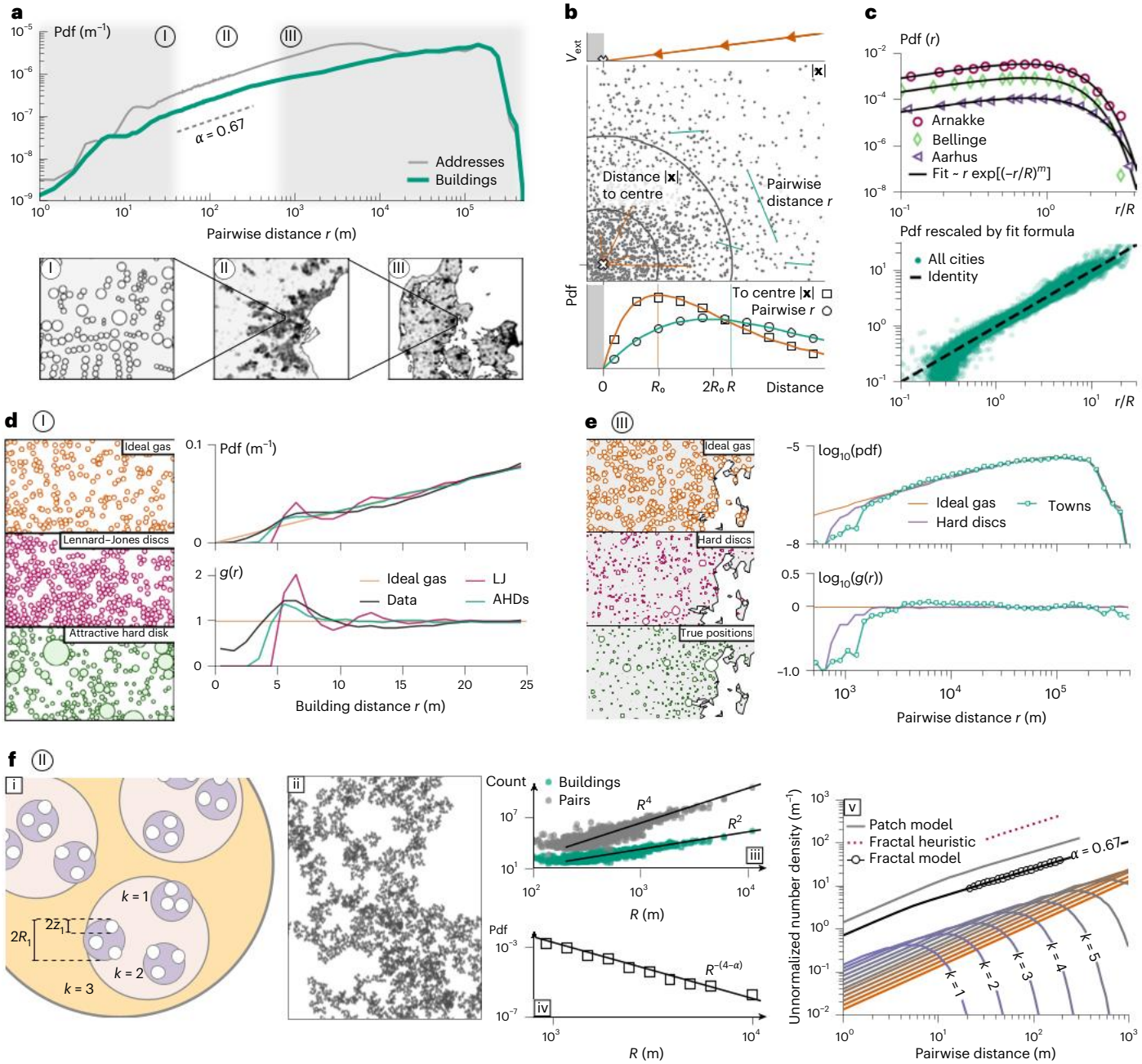


Fig. 3 | The structure of residential locations in Denmark. a, Pair distribution of residential locations at three scales: (I) the immediate neighbourhood, with a linear onset and an oscillating modulation; (II) the city scale, showing a power-law growth with exponent $\alpha = 0.67$; and (III) the country scale. **b**, With external potential $V_{ext}(|\mathbf{x}|)$, linear in distance to city centre $|\mathbf{x}|$, an ideal gas will form a city-like structure with radial density following an Erlang distribution of scale R_0 (squares). The corresponding pair distribution approximately assumes a generalized gamma distribution with scale R, providing a definition for city radius (circles). **c**, Maximum-likelihood fits of the generalized gamma distribution against pair distribution of buildings in three Danish towns (top) and all Danish towns, rescaled (bottom). **d**, Pair distributions for four systems of building positions on the micro-scale: (1) buildings in Denmark, (2) random positions (ideal gas), (3) Lennard–Jones discs (LJ) and (4) attractive hard discs (AHDs) of heterogeneous size. All systems display linear onset with oscillatory modulations. The pair-correlation function $g(r)$ quantifies the under- and over-presence of neighbours at distance r when compared with the ideal gas. Repulsive and attractive forces lead to valleys and peaks in $g(r)$. **e**, Pair distribution and $g(r)$ for all towns in Denmark: (1) random positions (ideal gas; thin, orange lines), (2) non-attractive hard discs (thick, violet lines) and (3) real data (dotted, teal lines). Systems 2 and 3 display similar behaviour with slow onset, quickly approaching the ideal gas. **f**, (i) Fractal model for building positions in cities. On each hierarchy layer, a patch of buildings consists of b sub-patches distributed with packing fraction θ. At the lowest level, a sub-patch is a building. (ii) Model sample with $b = 4$, $z_1 = 3$ m, $\theta = 0.8$ and $L = 8$. (iii,iv) Independent-patch model. Equating the concept of 'town' with 'patch of radius R containing buildings', the number of buildings per patch grows as R^2, and the number of pairs of buildings per patch grows as R^4 (with R from the analysis in **c**). The tail of the radius distribution decays as $R^{-(4-\alpha)}$ with $\alpha = 0.67$. (v) For both models as well as with heuristic $\alpha = 1 - 2\ln\theta/\ln(\theta/b) = 0.72$, we recover the pair distribution scaling observed in the data.

The condensed-matter physics of locations

Having illustrated the fundamental way geography is encoded in a country's location pair distribution and how it affects the dyadic process of human mobility, we now focus on a deeper analysis of its shape and origin. Specifically, we show how its properties naturally emerge from statistical physics arguments and how understanding the pair distribution allows us to identify key attributes of geography in terms of shaping human mobility. Thus, in this section, we go beyond simply

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appropriating the concept of the pair distribution function for analyses of human mobility and use the tools from condensed-matter physics[36] to create simple models for the micro-, meso- and macro-structure of the geography of an entire country.

The pair distribution between residential locations (buildings and addresses) in Denmark shows the behaviour displayed in Fig. 3a. In this view, buildings are simply addresses that stack on top of each other (for example, apartment buildings). On the micro-scale (I) within distances of $r \lesssim 25$ m, the 'immediate neighbourhood', we observe a linear onset of neighbour density, oscillatorily modulated. For larger distances at the meso-scale (II), this growth assumes a scaling of approximately $p(r) \propto r^{0.67}$ between ~25 m and ~200 m (until ~4,000 m for addresses). Afterwards, on the macro-scale (III), this growth curbs, decays rapidly and eventually approaches zero as we reach the limit of the finite system.

Regime I: city density as an ideal gas in a potential

Our basis for modelling cities is a simple particle model in two dimensions—where buildings are modelled as particles distributed uniformly at random, corresponding to an ideal gas kept in place by an external potential. The key constraint to identify the properties of this 2D confining potential (Fig. 3b) is the observation from the literature that population density per unit area decays exponentially with distance to its centre (Fig. 3c)[38–40]. A generalized gamma distribution of the shape $p(r) = (r/R^2)\exp[-(r/R)^m]/\Gamma(2/m)$ accurately describes the pair distribution of this ensemble, where the variable R allows us to associate every city with a radius that can be estimated from the data via a maximum-likelihood fit, without having to rely on a definition of city centre (Fig. 3c and 'City sizes' in the Supplementary Information).

To understand the oscillatory behaviour modulating the onset of the pair distribution, we took the condensed-matter approach a step further. We argue that there is both a repulsive and an attractive effective force between locations. The repulsive force originates from the simple fact that two buildings cannot occupy the same space. The effective attractive force results from the lower cost of placing buildings near one another as it reduces infrastructural cost for shared amenities[41].

In implementing this idea, we recognized that—unlike gases—cities are not constructed according to strict laws, so there is not a single, 'true' description of the system. We thus illustrated the influence of such effective forces on the pair distribution by modelling the system as two distinct ensembles of repulsive–attractive forces: (1) an ensemble of Lennard–Jones discs, a canonical model for sphere-like particles[42], and (2) attractive hard discs of heterogeneous size, both in the presence of a linear external potential (see 'Locations as interacting particles' in Methods). Both models reproduce the oscillatory features observed at the micro-scale in our high-resolution dataset (Fig. 3d).

We can study the oscillatory structure in terms of the pair-correlation function $g(r) = p(r)/i(r)$, where $i(r)$ is the pair distribution of an ideal gas. This function measures the over- and under-representation of neighbours at distance r in relation to the expectation if no interaction forces were present[43]. In both models and the data, there is a considerable lack of neighbouring buildings for distances $r \lesssim 4$ m, suggesting the presence of repulsive forces, later reaching a peak $g(r) > 1$ at $r \approx 6$ m, which is typically observed for systems with attractive forces. This peak becomes wider when we consider heterogeneous building radii. The discrepancy between the data and model(s) below 5 m (the distance between the black and teal/pink lines in Fig. 3d) is due to the definition of building location in the data as the location of its front door, rather than the building centre (see 'Data' in Methods).

Regime III: the spatial distribution of cities

We next used the condensed-matter toolset to understand the macro-scale (III) of the pair distribution. For large distances, its shape is dominated by the locations of cities. We compared an ideal gas of cities (that is, random positions uniformly distributed in the landmass of Denmark) to the empirical pair distribution of all Danish towns (Fig. 3e).

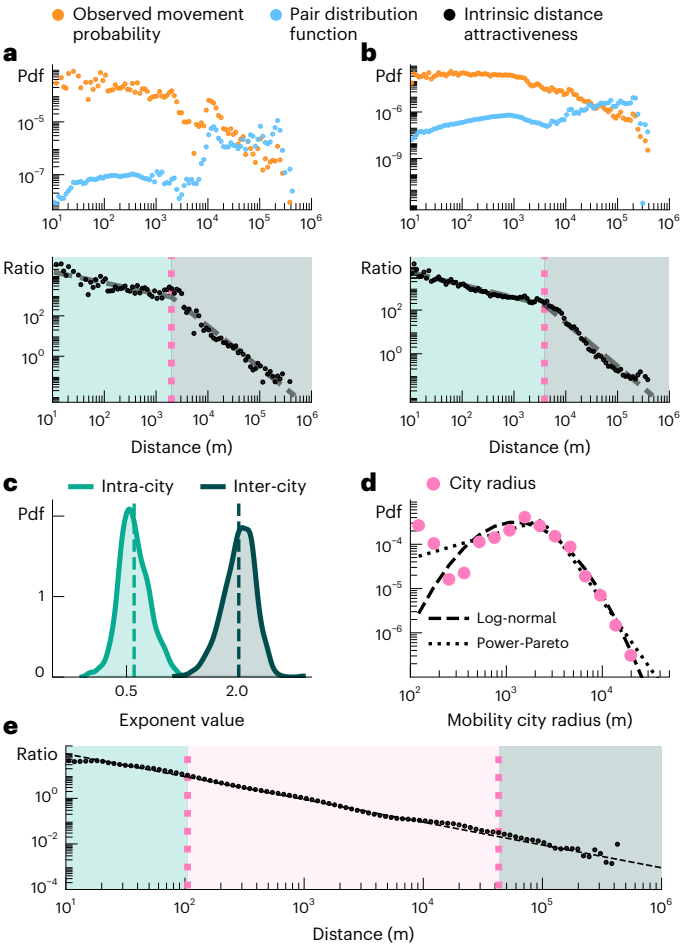


Fig. 4 | A piecewise process. a,**b**, Residential mobility analysis in a small (**a**) and a large (**b**) Danish town reveals a consistent piecewise power law across all cities. For more cities, see Supplementary Figs. 22–25; for details on the piecewise power-law definition and maximum-likelihood estimation, see Supplementary Information section 3.5. **c**, Distribution of the piecewise power laws' exponents. For the first and second parts, respectively, these centre on 0.6 ± 0.20 and 2.0 ± 0.21 (mean ± standard deviation). **d**, The transition point in the piecewise power law aligns with a city size distribution. We fit different distributions (log-normal and power-Pareto) via maximum likelihood (see 'Power-law estimation' in Methods). **e**, We recovered the intrinsic distance attractiveness from simulating the piecewise power law (**a**,**b**) with the mean values of 0.6 and 2 (**c**), over the city scale (**d**). We assumed random positions of cities as in Fig. 3e. For an alternative derivation of the intrinsic distance attractiveness and details, see Supplementary Information section 3.5.

While the ideal-gas distribution has a linear onset, there is a considerable lack of neighbouring towns for small distances $r \lesssim 1$ km, but after that, the observed town pair distribution rapidly approaches and matches the ideal-gas distribution remarkably well, demonstrated by the fast approach of $\log(g(r)) \to 0$.

The fact that there is not a clear peak in $g(r)$ suggests the absence of a strong attractive force. When we compared the observed pair distribution to that of a non-attractive hard-discs model (see 'City pair distribution and radius' in Methods), we observed a similar behaviour, indicating that the positions of towns are almost statistically indistinguishable from random locations except for the condition that towns may not overlap in territory. Next, we weighed every contributing pair distance for a pair of towns by the product of their respective numbers of buildings (Supplementary Information). While the hard-discs model aligns with the empirical observations for smaller distances, for larger distances the building-location pair distribution is only recovered by the empirical weighted city-location distribution—and not by the

hard-discs model. This is in line with observations that size plays a non-negligible role in where towns are located[44].

Regime II: fractal structures explain the meso-scale

As described above, the initial linear growth of the pair distribution quickly approaches a sub-linear scaling law of $\alpha \approx 0.67$ (Fig. 3a). The emergence of this scaling law is related to the non-regular shapes of cities composed of smaller 'patches' of buildings that form larger clusters in a space where occupation is limited by local geography (for example, bodies of water and hills) or for developmental reasons (for example, industrial areas, parks and agricultural use).

Specifically, we adapted a model that was recently used to generate surrogate city positions[32,45]. Assume that a city is constructed as a hierarchically nested structure of patches containing b buildings with packing fraction θ (Fig. 3f(i) and 'Fractal model' in Methods). For a range of realistic parameters, this model yields $\alpha = 0.67$ both for single realizations (Fig. 3f(ii)) and analytically. Deriving the heuristic approximation $\alpha = 1 - 2\ln\theta/\ln(\theta/b)$, we further showed that the scaling emerges independently from the choice of patch geometry (Supplementary Information section 2.7). Our model replicates the behaviour observed for the whole country and is in agreement with $\alpha = 0.69 \pm 0.04$ for individual cities. Furthermore, this nested description is consistent with earlier observations that settlement size distributions are scale-free[22,46].

Additionally, by modelling the built environment as a multiverse of independent patches and inferring the patches' area distribution from the distributions of radii using the per-city pair distributions (Fig. 3f(iii) and 'Patch model' in Methods), we obtained $\alpha = 0.71$ as well as a patch area distribution of $f(A) \propto 1/A^{2.15}$, a result that matches the exponents reported in ref. 46. The meso-scale scaling law is robust, as all these similar values of α arising from different models are consistent with our empirical finding (Fig. 2a and Extended Data Fig. 1) and the literature[46].

From local to global power laws

Considering the intrinsic distance attractiveness starting from individual Danish cities (as opposed to globally), we observed a piecewise process (Supplementary Figs. 22–26)[47]. Specifically, for each city, we considered the distribution of moves originating in the city of interest to the rest of the country. The pair distribution is limited to pairs that include at least one address in the city of interest. At the city scale, we found that the intrinsic distance attractiveness is a piecewise power law (Fig. 4a,b) with the first exponent centred on 0.60 ± 0.20 (s.d.) and the second exponent on 2.0 ± 0.21 (s.d.) (Fig. 4c). The process is universal in the sense that the exponents of the piecewise power laws are consistent across the 1,400 cities of Denmark. The transition point between the two power laws defines a mobility city radius (Fig. 4d) with a distribution that lies between a log-normal and a power-Pareto distribution (see 'Power-law estimation' in Methods).

Now we could put the pieces together. We did this by exploring how the global power law (Fig. 2b) relates to the local piecewise picture for each city (Fig. 4a–b). Here, our statistical physics framework describing the components of the pair distribution could be used to complete the picture of human mobility. Finally, we could reproduce the intrinsic distance attractiveness starting from the city-level piecewise description. We simulated the piecewise power law over a toy geography that reproduces the key characteristics of geography: scale of cities (Fig. 4e), local pair distribution that follows a generalized gamma distribution (Fig. 3c and equation (17) in the Supplementary Information) and random positions of cities (Fig. 3e(iii)). We recovered the empirical intrinsic distance attractiveness, $\pi(r) = 1/r$, validating our geographical model and the piecewise power law (Supplementary Information section 3.5).

Discussion

Geography trivially constrains human mobility. While the existing human mobility literature has extensively documented the structure of human mobility, it often remains disconnected from studies focused on the organization of the built environment. This has led to the emergence of two distinct perspectives on human mobility[24]: one based on distance, similar to the gravity model[12], and another focused on the availability of opportunities[27]. Here we have proposed the pair distribution of locations as a key normalization variable of human mobility. The normalization reveals an intrinsic power law consistent across various geographies and mobility types. Additionally, we proposed a model inspired by statistical physics to account for the shape of the pair distribution between locations. A limitation of our work is that while the model replicates pair distribution characteristics for general addresses at dwelling and country scales, the analysis does not yet extend to the pair distribution between non-residential of locations, such as home and work pairs. One could, for example, compute the pair distance function for all possible ('home', 'work') location pairs. The resulting empirical 'home'–'work' trip distance distribution would capture the spatial constraints on travel patterns. Then, using the approach outlined in the manuscript, dividing the commute trip distance distribution by the ('home', 'work') pair distance distribution, we can reveal an intrinsic distance pattern for this type of activity. Such an extension could help us understand commuting patterns, a significant component of day-to-day mobility[48–50]. The temporal scope of the analysis is also a limitation. We considered the period from 1986 to 2020 as an aggregated timescale. Our study therefore captures long-term average characteristics but does not resolve dynamics or evolutionary trends within this 34-year span, nor does it account for the impact of specific events or changes during these decades. Furthermore, the fact that we do not have data from periods prior to 1986 prevents a study of historical changes of the observed mobility structures. This restricts our ability to fully understand how these patterns have evolved from earlier states of the built environment and societal organization, or to ascertain whether the observed power law is a long-standing feature or one that emerged recently. Finally, we explained our normalization as a continuous gravity model that reconciles the two paradigms of human mobility. We also show that this description does not hold from the perspective of individual cities and emerges from the aggregation of scales[15]. Moreover, our approach provides a baseline in the form of the intrinsic distance attractiveness, $\pi(r) = 1/r$, which serves as a null model for exploring mobility patterns of sub-populations. Exploring deviations from this baseline across different demographic groups—such as those based on age, gender or socio-economic status—could offer valuable insights into how specific factors shape human movement. Incorporating such new demographic insights could deepen our understanding of the underlying variability in mobility patterns across different population segments. At the city level, other demographic factors could explain relocation decisions, such as housing prices, employment opportunities and family evolution[47]. Our framework expands the understanding of human mobility and provides a bridge connecting two rich bodies of literature.

Methods

Data

Uncertainty quantification. Estimates reported in the main text are reported as mean ± standard deviation.

Residential mobility in Denmark. The data on residential mobility in Denmark are derived from the Befolkning database of Danmarks Statistik[51]. This database contains 39,297,646 residential moves between different addresses in Denmark from 1986 to 2020. The location data for the 2,345,453 buildings that make up the 3,251,464 addresses are also provided by Danmarks Statistik, with an accuracy better than ±2 metres[52]. The location of an address is defined as the location of the building's entrance door; we investigate the effect of this definition of the pair distribution function in Supplementary Information section 2.4. These datasets are not publicly available due to Danish Data

Protection regulations. However, access to the data is possible for research purposes. The data can be obtained via Statistics Denmark for Researchers in accordance with Statistics Denmark's Research Scheme: www.dst.dk/en/TilSalg/Forskningsservice/.

Bias in residential mobility in Denmark. In contrast to studies using digital tools such as mobile phones, which may introduce biases[53], the Danish residential mobility dataset we study is inherently representative of the entire population. In fact, the dataset includes every residential move, a requirement for all residents to report, thus eliminating potential biases associated with selective data collection methods[54]. Demographic information (gender and year of birth) is reported in Extended Data Figs. 2 and 3 (refs. 55,56). Cities are defined according to the definition of Danmarks Statistik; to ensure the robustness of the results, we compared their definition with a hierarchical clustering of the addresses (HDBSCAN[57] in Supplementary Information section 4).

Residential mobility in France. The location data for France are publicly available and were compiled from the Base Adresse Nationale, accessible via https://adresse.data.gouv.fr/donnees-nationales, which contains precise location data for 21,567,447 buildings. To match the housing unit data, we used information from https://www.data.gouv.fr (FiLoSoFi), which segments the country into a grid of 200-m squares, each annotated with the count of housing units and collective housing entities. We augmented the data by uniformly distributing the housing unit count among each collective housing entity, resulting in 34,041,910 individual addresses. The residential mobility data are publicly available and were obtained from https://www.insee.fr/fr/statistiques. They contain 40,465,288 inter-city migrations from 2016 to 2020. The distances are calculated from one city centre to another. The city centres are computed as the centroid of contained addresses. As we only had access to inter-city moves, migration distances that are less than the largest city's dimensions in France were disregarded.

Day-to-day mobility. The data on daily mobility were obtained from user check-ins (points of interest) on the location-centric social network Foursquare over six months (27 May 2010 to 3 November 2010) as previously documented[28,58]. The Foursquare dataset was sourced entirely from publicly available data: check-ins were accessed via Twitter's Streaming API and venue geo-coordinates from the Foursquare website, both being public at the time of collection. User consent was implied by their public sharing of this information on these platforms, and data access was therefore governed by the terms of these public channels. The data comprise a total of 239,788 movements across 43,395 distinct locations. The regional breakdown of the data is 47,996 movements across 11,808 locations in Houston, 79,624 movements across 15,617 locations in Singapore and 112,168 movements across 15,970 locations in San Francisco.

GeoBoundaries. We obtained the country, state and municipality shapes from the geoBoundaries project[59]. Throughout the paper, we worked with the EPSG:23032 projection for all points that lie within Denmark and the EPSG:27561 projection for points that lie within France. We then performed analyses on the Euclidean geometry of the projection.

Pair distribution function

In this section, we provide a formal definition of the pair distribution function. We consider points $x_i \in \Omega$ distributed in a d-dimensional space $\Omega \subseteq \mathbb{R}^d$ according to a density $\varrho^d(x)$; then, the pair distribution $p(r)$ is unidimensional and determined by both the shape of the space and $\varrho^d(x)$. The pair distribution is thus given by

$$p(r) = \int_\Omega \mathrm{d}x \int_\Omega \mathrm{d}y\, \varrho^d(x)\varrho^d(y)\delta(r - \| x - y \|). \tag{4}$$

where $\|x-y\|$ is the distance between two points x and y. When we analyse flows between pairs of points in Ω, we notice a distribution that depends on the distance, referred to as $f(r)$, or the observed movement distance. The anticipated number of movements from a location x to a location y is determined by this distribution:

$$f(x,y)\mathrm{d}^2x\mathrm{d}^2y \tag{5}$$

where $f(x,y)$ is a continuous number density. This number is influenced by two functions. First, the number of observations is proportional to the number of locations at x as well as the number of locations at y—that is, depending on the product of location density $\varrho(x)\varrho(y)\mathrm{d}x\mathrm{d}y$. Second, the number of movements is proportional to the movement propensity—that is, the tendency for a move that begins at a location at x to end at a location at y, which we denote as $\pi(x,y)$:

$$f(x,y)\mathrm{d}^2x\mathrm{d}^2y = \pi(x,y)\varrho^d(x)\varrho^d(y)\mathrm{d}^2x\mathrm{d}^2y. \tag{6}$$

The number of moves spanning distance r within the domain Ω is

$$f(r) = \int_\Omega \pi(x,y)\varrho^d(x)\varrho^d(y)\delta(r - \| x - y \|)\mathrm{d}x\mathrm{d}y. \tag{7}$$

If we make the anstaz that the propensity to travel from x to y solely depends on the distance between the two; that is:

$$\pi(x,y) = \pi(\| x - y \|) = \pi(r) \tag{8}$$

as $\pi(r)$ is now independent of location x and y, it can be factored out of the integral, and we obtain

$$f(r) \propto \pi(r)p(r) \tag{9}$$

where $p(r)$ is the pair distribution function as in equation (4), and we recover the main text's equation (1) (see Supplementary Information section 1.2 for more details). The ansatz of equation (8) is the key point to reconcile the distance-based and opportunity-based perspectives on human mobility. The distance term is $\pi(\|x-y\|)$, and the opportunity term is the expected number of pairs of locations between location x and location y, $\varrho^d(x)\varrho^d(y)\mathrm{d}^2x\mathrm{d}^2y$ (Supplementary Information sections 1.2 and 3.3)[60,61].

Moreover, the pair distribution function characterizes the geometry of the space as it represents the joint probability of finding two locations at particular positions in the system (Supplementary Information sections 1.1 and 1.5). The computation of the pair distribution function scales as $\mathcal{O}(n^2)$; computing it for a large number of locations is challenging (Supplementary Information section 2.5).

Locations as interacting particles

Our basis for modelling cities is a simple particle model in two dimensions—where buildings are modelled as particles uniformly distributed at random, corresponding to an ideal gas kept in place by an external potential. Consider the city centre (for example, the central business district) as the primary point of interest due to its proximity to amenities. This attractiveness implies a higher cost of being located further away from the centre, represented by the distance $|\mathbf{x}|$.

At the same time, two simple mechanisms will prevent buildings from accumulating in the exact centre of the city. First, buildings have a certain average radius z, so they cannot be too close (at distance $<2z$). There is an advantage to buildings not being too far apart, as they can share local amenities. Rather than explicitly modelling this phenomenon, we assume that there is an inherent temperature, T, in the system. This temperature determines the distribution of house locations, which is influenced by interactions and external potential.

From a statistical physics perspective, we describe the system with a simple external potential that linearly increases with distance:

$$V^{\text{ext}}(\boldsymbol{x}) = \gamma|\boldsymbol{x}| \tag{10}$$

where we assume that the origin of the coordinate system is located at the city centre. To model repulsion and attraction between houses, we further assume a Lennard–Jones interaction potential

$$V^{\text{int}}(\mathbf{x}_i, \mathbf{x}_j) = \varepsilon\left[\left(\frac{2z}{|\mathbf{x}_j - \mathbf{x}_i|}\right)^{12} - 2\left(\frac{2z}{|\mathbf{x}_j - \mathbf{x}_i|}\right)^{6}\right], \tag{11}$$

which is commonly used to model intermolecular attraction and repulsion[36,42].

In total, a system with these properties evolves according to the Hamiltonian

$$\mathcal{H}(\{\mathbf{x}, \mathbf{p}\}) = \frac{1}{2}\sum_{i=1}^{N}\mathbf{p}_i^2 + \gamma\sum_{i=0}^{N}|\mathbf{x}_i| + \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i}^{N}V^{\text{int}}(\mathbf{x}_i, \mathbf{x}_j) \tag{12}$$

with two-dimensional momenta $\mathbf{p}_i$ and locations $\mathbf{x}_i$. In a canonical-ensemble formulation of the system that is at constant inverse temperature β, the probability of finding a configuration $\{\mathbf{x}, \mathbf{p}\}$ in volume-element $\mathrm{d}^2\mathbf{x}\mathrm{d}^2\mathbf{p}$ is given by

$$\varrho[\{\mathbf{x}, \mathbf{p}\}]\mathrm{d}^2\mathbf{x}\mathrm{d}^2\mathbf{p} = \exp[-\beta\mathcal{H}(\{\mathbf{x}, \mathbf{p}\})]\mathrm{d}^2\mathbf{x}\mathrm{d}^2\mathbf{p}. \tag{13}$$

We first consider an ideal gas where $V^{\text{int}} = 0$, which enables us to say something about the density of particles around the centre of city—that is, we want to find the particle dwelling probability $p(r)dr$, where r is the particle's distance to the centre. For the sake of simplicity, we set $N = 1$ without loss of generality. This is justified by the ergodicity of the system, which implies that the trajectory of a single particle will eventually follow the density of the entire distribution. This can be expressed as taking the Nth root of the N-particle density. Integrating over the momenta yields

$$\varrho[\mathbf{x}]\mathrm{d}^2\mathbf{x} = \exp\left(-\beta\gamma|\mathbf{x}|\right)\mathrm{d}^2\mathbf{x}. \tag{14}$$

With a change of variables to polar coordinates, we find

$$\varrho[r, \phi]r\mathrm{d}r\mathrm{d}\phi = r\exp\left[-\beta\gamma r\right]\mathrm{d}r\mathrm{d}\phi, \tag{15}$$

that is,

$$p(r)\mathrm{d}r = r\exp\left[-\beta\gamma r\right]\mathrm{d}r. \tag{16}$$

This implies that in the absence of interactions, the distribution of houses around the city centre should follow a $k = 2$ Erlang distribution with a scale parameter of $\lambda = \beta\gamma = 1/R_0$, where R_0 is half the city radius. The role of the inverse temperature term, β, is explained in Supplementary Information section 2.1.4.

To gain insight into the radial particle density in the context of strong repulsion, we return to the Lennard–Jones perspective. In the limit of $\varepsilon/T \gg 1$, particles will have a strong tendency to be found in their respective potential minimum—that is, at distance $2z$ from each other. Effectively, we can think of them as hard discs of radius z that tend to form clusters. If the temperature is low, the effective radius of the city will be small, and we expect a crystal to form in the centre (Supplementary Information section 2.1.4).

Molecular dynamics simulation. We are interested in finding configurations $\mathcal{C}$ that accurately depict the canonical ensemble with number of particles N, an average-constant temperature T and a constant but irrelevant volume. We assume that the particles are constrained to a radially symmetric external potential, with the total volume of the system containing the particles being irrelevant. Finally, the total potential must be

$$V = \sum_{i=1}^{N}\left[V^{\text{ext}}(\mathbf{x}_i) + \frac{1}{2}\sum_{j\neq i}V^{\text{int}}(\mathbf{x}_i, \mathbf{x}_j)\right]. \tag{17}$$

To this end, we integrate the equations determined by the system's Hamiltonian numerically using the velocity-Verlet algorithm[62]. Furthermore, we rescale particle velocities according to the stochastic Berendsen thermostat with relaxation time τ (ref. 63). The details and parameters of the molecular dynamics simulation are available in Supplementary Information section 2.1.5.

Hard discs. We generate a configuration of hard discs with a non-specified attractive force (that is, we do not explicitly integrate the equations of motion for a hard-sphere interaction potential with an additional attractive force). To do so, we first generate $N = 10^4$ random positions according to a radial Erlang distribution of scale parameter $R = 675$ (that is, an ideal gas). Subsequently, we draw a random radius for each position from a heterogeneous distribution with power law tail. We first draw values $\hat{z}_i$:

$$p(\hat{z}) = \frac{a^2 - 1}{2a} \times \begin{cases} \hat{z}^a, & \hat{z} \leq 1, \\ 1/\hat{z}^a, & \hat{z} > 1, \end{cases} \tag{18}$$

and then assign disc radius $z_i = 3\hat{z}_i/ < \hat{z} >$. We choose $a = 4$ to obtain a heterogeneous distribution with non-finite variance in disc area. Afterwards, we run the collision algorithm outlined in Supplementary Information section 2.2.

The modelling of interaction between the house in the external potential is further developed in Supplementary Information section 2.3.

Non-overlapping discs

To emulate the positions of cities, we randomly distribute N hard discs of radii R_i in shape Ω. We start with the largest disc of radius $R_{\max}$ and iterate over all discs, ordered decreasingly in size. For every disc j, we generate a random position $\mathbf{x} \in \Omega$ until the condition $|\mathbf{x} - \mathbf{x}_i| > R_j + R_i, \forall j < i$ is met—that is, drawing new random positions until there are no overlaps with other, already placed discs. Then, we assign $\mathbf{x}_j \leftarrow \mathbf{x}$ and continue with the next disc.

City pair distribution and radius

The pair distribution of each city is best represented by a generalized gamma distribution with linear onset

$$p(r, R, m) = \frac{r}{R^2\,\Gamma(2/m)}\exp\left[-\left(\frac{r}{R}\right)^m\right], \tag{19}$$

as depicted in Fig. 3c. Here, r is the distance between two locations, R is the scale parameter corresponding to the city radius and $m > 0$ is a shape parameter controlling the decay of the tail. Assuming that population density decays exponentially with distance to the city centre and that consequently the pair distribution of buildings assumes a generalized pair distribution of the form of equation (19), we perform maximum-likelihood fits to infer parameters R and m for each city in Denmark with more than 30 buildings. We generally find strong correspondence to the empirical distribution, as illustrated by the correlation of r/R and the inverse of equation (19):

$$(r/R)^* = -(\log[p(r)R^2\Gamma(2/m)/r])^{1/m}, \tag{20}$$

albeit with small deviations at smaller distances (Fig. 3c) that can be associated with the sub-linear scaling behaviour observed on the

meso-scale of Fig. 3a. Remarkably, measuring the 'radius' R of a city in terms of its pair distribution's scaling parameter comes with the advantage of not having to rely on any definition of 'city centre'.

Moreover, no assumptions about urban growth are necessary; instead, the observed behaviour emerges from the simple ansatz that being placed at distance $|\mathbf{x}|$ from the centre of a town comes at a cost $\propto |\mathbf{x}|$, which can be overcome for a multitude of reasons encoded in temperature T.

Fractal model. The fractal model we adopt was originally devised to explain the positions of galaxies and was recently used to generate surrogate city positions[32,45]. We assume that a city is constructed as a hierarchically nested structure of patches containing buildings (Fig. 3f(i) and Supplementary Information section 2.7). At layer $k = 1$, a patch of radius R_1 contains b buildings of radius z_1 placed in a non-overlapping manner with packing fraction θ. At layer $k = 2$, a patch of radius R_2 contains b building patches of radius $z_2 = R_1$, again placed without overlap and packing fraction $\theta = \theta_k = bz_k^2/R_k^2$. Continue in that fashion until reaching layer $k = L$. Due to the constant packing fraction $\theta < 1$, larger areas of absent buildings will form at each hierarchy layer, mirroring similar observations in real urban structures. Note that demanding θ to be constant across hierarchy layers also implies a predetermined patch radius of $R_k = (\theta/b)^{k/2} z_1$. A sample configuration with $b = 4$, $z_1 = 4$ m and $\theta = 0.8$ can be seen in Fig. 3f(ii).

To evaluate the overall pair distribution of such a fractal structure, we assume that for a patch of hierarchy layer k, the b^{k-1} buildings that are contained in each of its b sub-patches are all located at their respective centres. Then, the pair distance number density (pdnd) of an average such patch will approximately assume equation (19), weighted with a total of $b^{2k-2}b(b-1)/2$ building pairs (between subpatches). Moreover, at every hierarchy layer k there will be b^{L-k} such patches. In total, each hierarchy layer therefore contributes a pdnd of

$$n_k(r) \propto b^k p(r, R_k, m), \tag{21}$$

implying that the observed pdnd of the whole structure is proportional to

$$n(r) = \sum_{k=1}^{L} n_k(r) \propto \sum_{k=1}^{L} \frac{b^{2k} r}{\theta^k z_1^2} \exp\left[-\left(\frac{r}{z_1}\right)^m \left(\frac{b}{\theta}\right)^{mk/2}\right]. \tag{22}$$

In the limit of $L \to \infty$, we use Laplace's method to find

$$\log n(r) \propto \underbrace{\left(1 - \frac{2\log\theta}{\log(\theta/b)}\right)}_{=\alpha} \log r, \tag{23}$$

that is, a sub-linear scaling law emerges from considering the pair-weighted sum of single-scale pair distributions with linear onset. The pdnd $n(r) \propto r^\alpha$ with $\alpha = 1 - 2\log\theta/\log(\theta/b)$. Notably, the result is independent of the parameter m that controls the tail of the single-scale pair distribution, suggesting that the exact shape of the composite pair distributions is irrelevant, which we further demonstrate in Supplementary Information section 2.1 by discussing another functional form of the pair distribution.

Patch model. We consider a multiverse consisting of an infinite amount of universes indexed i, each of which is inhabited by a patch of size R_i with N_i buildings inside, contributing to the multiverse with a pair distribution number density of $n(r, R_i) \approx N_i^2 p(r, R_i, m)$ (each universe contributes a number of building pairs amounting to N_i^2). We also assume that there is a constant population density $\rho_0 \propto N/R^2$, demanding that $N \propto R^2$.

For each of these universes (or rather, each of these patches), we calculate the number of building pairs at distance r and assume that patch sizes R are distributed according to some—at this time still unknown—distribution $f(R)$. Then, the joint distribution of house pairs at distance r is given by

$$n(r) \propto \int_0^\infty \mathrm{d}R\, n(r, R) f(R) \tag{24}$$

$$\propto \int_0^\infty \mathrm{d}R f(R) R^4 \frac{r}{R^2} \exp\left(-\left(\frac{r}{R}\right)^m\right). \tag{25}$$

To simplify the integral, we change variables to $\beta = 1/R$, such that $\mathrm{d}R = -\beta^2 \mathrm{d}\beta$ and

$$n(r) \propto \int_0^\infty \mathrm{d}\beta f(\beta^{-1}) \frac{r}{\beta^4} \exp\left(-(\beta r)^m\right). \tag{26}$$

This integral yields a solution if $f = \beta^{4-\alpha}$ with $\alpha < 1$. Let us assume that this is the case, which means we have to solve the integral

$$n(r) = C \int_0^\infty \beta^{-\alpha} r \exp\left(-(r\beta)^m\right) \mathrm{d}\beta \tag{27}$$

where C is a normalization constant. We recognize the integral of the gamma function (Supplementary Information section 2.6) and therefore find

$$n(r) \propto r^\alpha C \Gamma\left(\frac{1-\alpha}{m}\right). \tag{28}$$

That is, if the radius of patches in the multiverse would be distributed as $f(R) \propto R^{-(4-\alpha)}$, the initial growth of the joint pair distribution would follow a sub-linear power law. Notably, the scaling exponent does not depend on the tail parameter m.

Let us see what this would mean in terms of the area of the patches. The area would scale as $A \propto R^2$ (or $R \propto \sqrt{A}$); that is, the distribution of the area would follow

$$\tilde{f}(A) = \frac{\mathrm{d}R}{\mathrm{d}A} f\left(R = \sqrt{A}\right) \tag{29}$$

$$\propto \frac{1}{A^{(5-\alpha)/2}}. \tag{30}$$

In the data, we observed $\alpha = 0.67$. That would mean that the area of the patches would have to be distributed according to a power law with exponent $\mu = -(5-\alpha)/2 \approx -2.17$, which is well within what has been found empirically for cities[46]. Furthermore, we found $f(R) \propto 1/R^{3.29}$ from our city radius inference analysis (Supplementary Fig. 11), which leads to $\mu = -2.15$, showing that the results of these two separate analyses are consistent.

A 'continuous' gravity law

In its original form, the gravity law for human migration states that the probability of moving between two cities is proportional to the product of their population and inversely proportional to the distance separating them[12]:

$$P_{1\to 2} \propto \frac{N_1 N_2}{d_{1\to 2}}. \tag{31}$$

This relationship requires administrative units to define cities and their respective populations, thereby rendering it inherently discrete

and subject to arbitrary choices regarding administrative boundaries. Yet, constructing a product of origin × destination populations closely aligns with the idea of normalizing by the pair distribution. To illustrate, consider two cities of populations N_1 and N_2, with radii r_1 and r_2 separated by a distance d. If we assume a uniform location density within these cities (as illustrated in Fig. 2c), the density of locations is given by

$$p^d(x) = \frac{N_1}{\pi r_1^2}\delta_{\Omega_1}(x) + \frac{N_2}{\pi r_2^2}\delta_{\Omega_2}(x), \tag{32}$$

where Ω_j is the domain of city j. This leads to the pair distribution (using equation (4)) :

$$p(r) = N_1N_2\int_{\Omega}\int_{\Omega} \mathrm{d}x\mathrm{d}y \frac{1}{\pi r_1^2}\frac{1}{\pi r_2^2}\delta_{\Omega_i}(x)\delta_{\Omega_j}(y)\delta(r - R(x,y)). \tag{33}$$

In the limit case where d significantly exceeds both r_1 and r_2, when the city radii are negligible compared with the distance between cities, the pair distribution around d tends towards a Dirac distribution of height equal to the product of the number of addresses in each city, $p(r) \underset{d\gg r_1,r_2}{\to} N_1N_2\delta_d(r)$; therefore (Extended Data Fig. 4),

$$f(r) = p(r)\pi(r) \to \frac{N_1N_2\delta_d(r)}{r}. \tag{34}$$

Equation (34) illustrates how in the context of the gravity model for migrations, the 'mass-product' N_iN_j emerges. Given that the number of addresses in a city is proportional to its population, we recover the discrete gravity law. We refer to our finding as 'continuous' since it operates at the granularity of individual addresses, the finest possible scale (Supplementary Information section 3.1.1). A comparison of the gravity model, the radiation model and the pair distribution framework is shown in Extended Data Fig. 5.

We can also interpret the intrinsic distance attractiveness $\pi(r) = 1/r$ as an emergent property of random utility theory, where individuals appreciate distance with a logarithmic scale. This interpretation is consistent with the logarithmic utility function described in Supplementary Information section 3.2.

Power-law estimation

The parameters for power-law distributions are determined using maximum-likelihood estimation, as outlined in ref. 64. For continuous distributions, the probability density function for a power-law distribution is given by

$$p(x) = Cx^{-\alpha} \tag{35}$$

where $x > x_{\min} > 0$ and C is a normalization constant. The parameter α is typically greater than 1, as the probability density function described in equation (35) does not integrate over $x \in [x_{\min}, +\infty)$ for $\alpha \leq 1$. However, in our analysis, the intrinsic distance attractiveness has an exponent $\alpha \approx 1$. Moreover, the maximum-likelihood estimation method from Clauset et al.[64] is known to be biased when $\alpha < 1.5$, a limitation that has been confirmed by Hanel et al.[65]. Consequently, to estimate power laws with $\alpha < 1.5$, we employ the maximum-likelihood estimator developed in refs. 65,66, with a maximum boundary for the power law, so that the probability density function in equation (35) is integrable over $x \in [x_{\min}, x_{\max}]$ for any $\alpha > 0$. It is reasonable to assume that power-law distributions are bounded within the context of this study, given that geographical limitations inherently bound the movement distances in our dataset. We define $x_{\max}$ as the diameter of each domain Ω, which corresponds to the largest distance between two points (Extended Data Table 1). We assess the power-law parameter estimates through goodness-of-fit tests employing the Kolmogorov–Smirnov statistic and likelihood ratios. The Kolmogorov–Smirnov tests are presented with P values in Extended Data Table 1. Likelihood ratios provide evidence on whether the power-law distribution is a superior fit compared with other distributions. The methodology for estimating piecewise power laws for each Danish city (Fig. 4a,b) is based on a maximum-likelihood estimator derived in ref. 67, and the details of the fit are available in Supplementary Information sections 3.4 and 5. In the analysis of the mobility city radius, the power-Pareto distribution[67] is identified as the optimal model with a slightly better fit, measured by an Akaike information criterion score of 24,046.03. The log-normal distribution has an Akaike information criterion of 24,161.17. A comparison with additional distributions is available in Extended Data Table 2.

The meso-scale power law of the pair distribution function (Fig. 3a) and the simulation of the fractal and patch model (Fig. 3f(v)) are also estimated with a maximum-likelihood estimator.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data for Denmark used in this study are not publicly available due to Danish Data Protection regulations. However, access to the data is possible for research purposes. The data can be obtained via *Statistics Denmark for Researchers* in accordance with Statistics Denmark's Research Scheme: https://www.dst.dk/en/TilSalg/Forskningsservice/. The location data for France are publicly available at https://adresse.data.gouv.fr/donnees-nationales and https://www.data.gouv.fr. The residential mobility data were obtained from https://www.insee.fr/fr/statistiques. The Foursquare data are publicly available at https://sites.google.com/site/yangdingqi/home/foursquare-dataset. Source data are provided with this paper.

Code availability

The geography model and estimation code are available via GitHub at https://github.com/benmaier/ljhouses. The code for the figures, data and statistical analysis is available via GitHub at https://github.com/LCB0B/role-of-geo. The code is also available via Zenodo at https://doi.org/10.5281/zenodo.14329837 (ref. 68).

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Author contributions

L.B., B.F.M. and S.L. designed the study and the model. L.B. and B.F.M. performed the analyses and implemented the model. L.B., B.F.M. and S.L. analysed the results and wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

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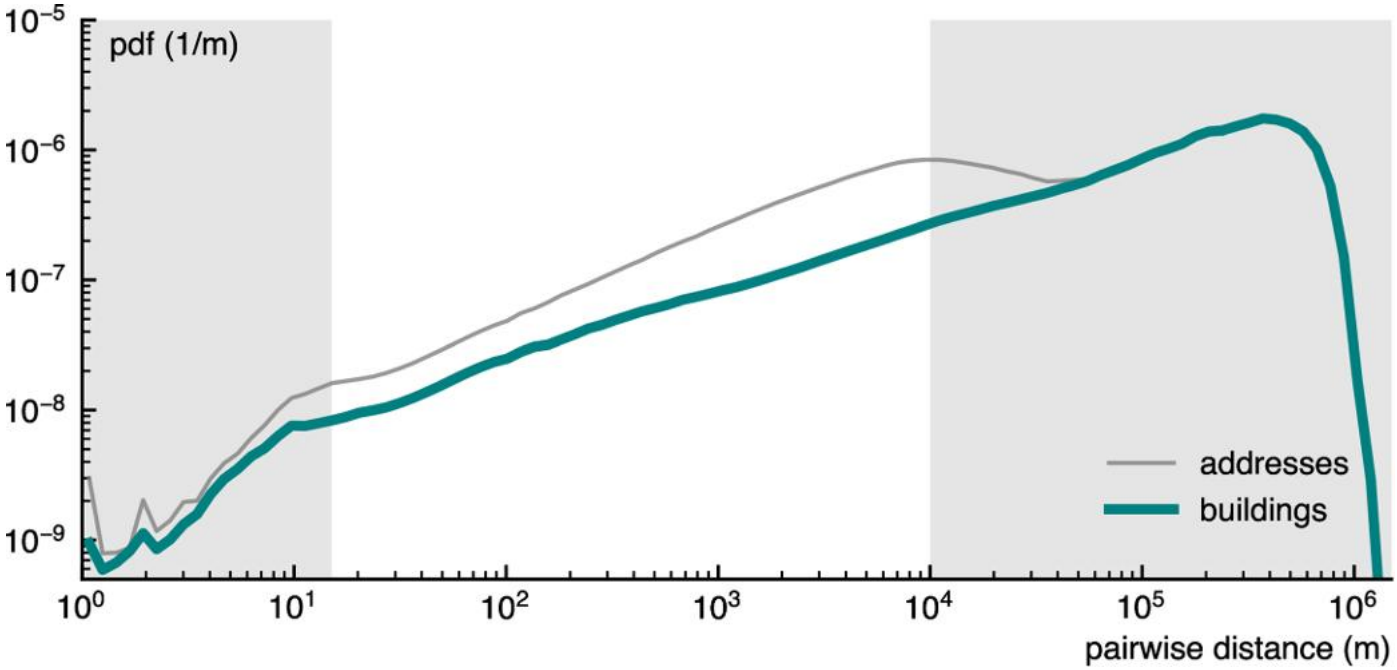
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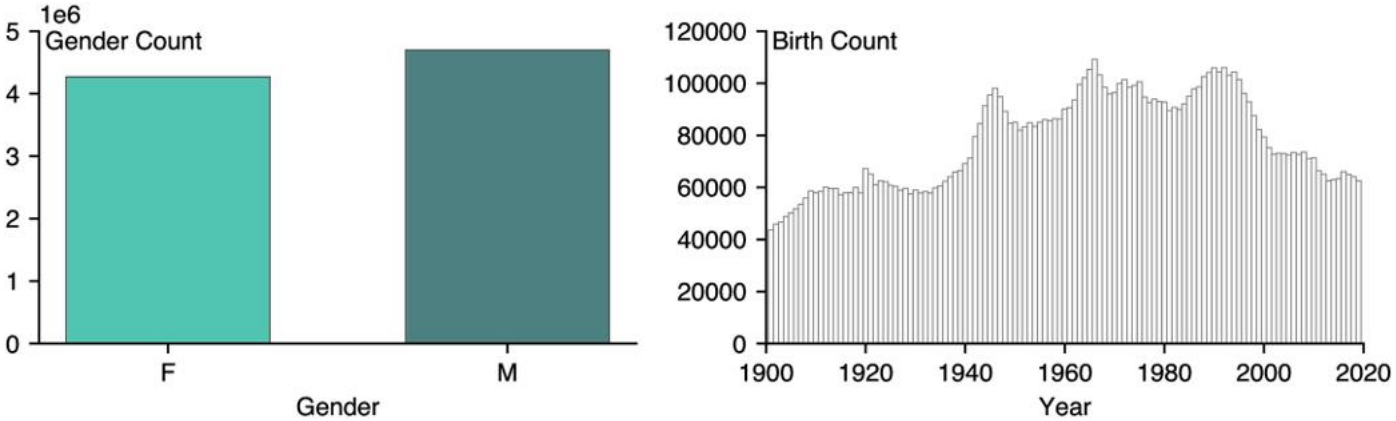
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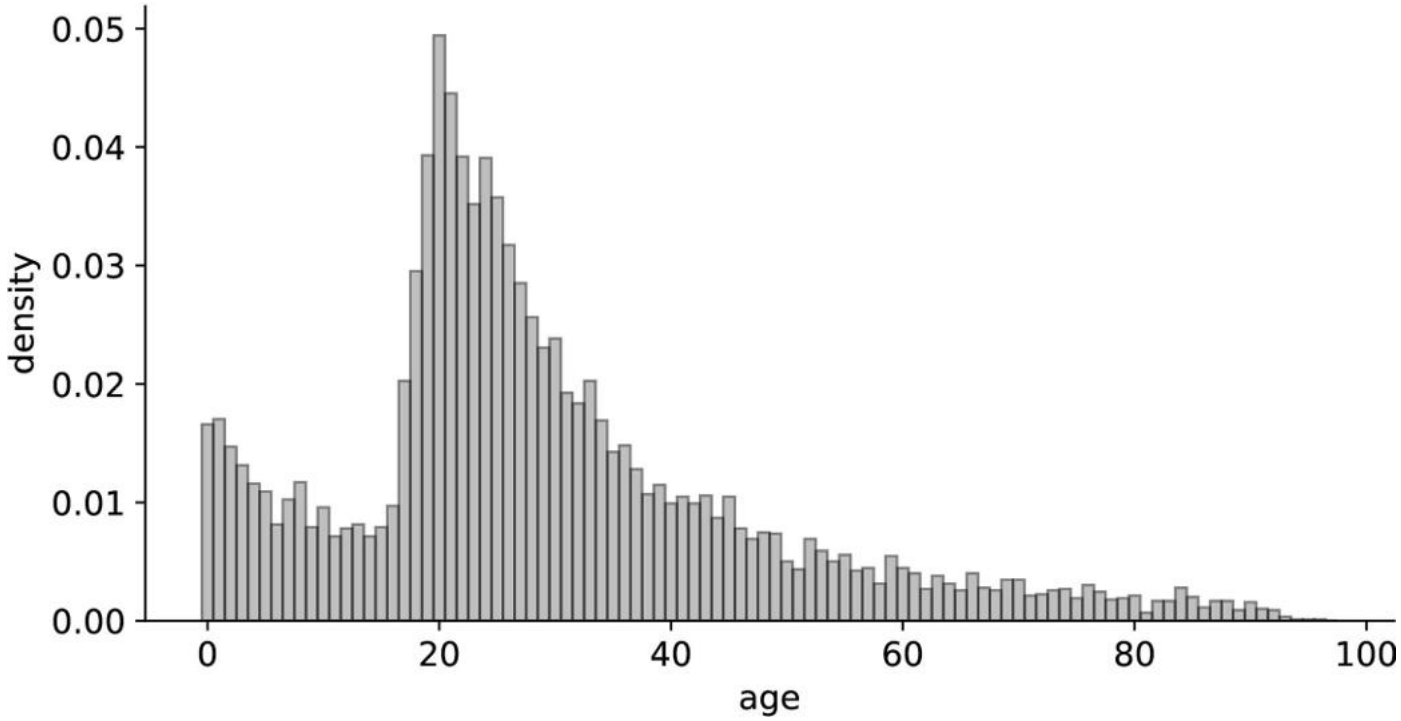
Extended Data Fig. 1 | Pair distribution of buildings and addresses in France. Pair distribution of buildings and addresses in France shows three scales: (I) Immediate neighborhood-scale with linear onset and oscillating modulation, (II) city-scale with power-law growth ($\alpha = 0.67$, for addresses), and (III) country-scale with slower growth and rapid decay (see Supplementary Section 1.6).

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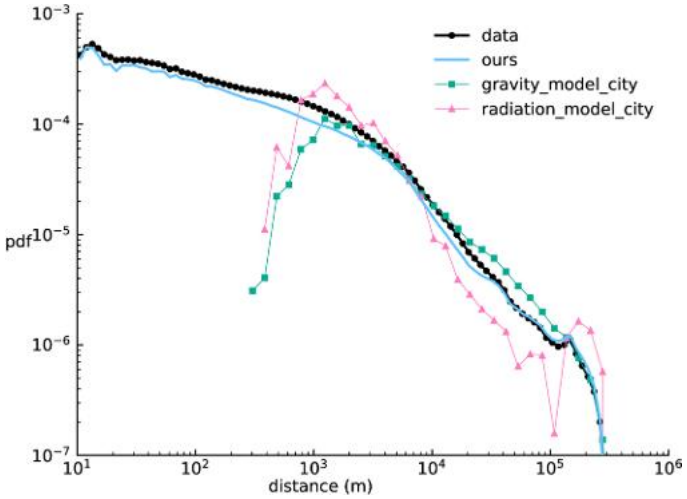
Extended Data Fig. 2 | Demographic characteristics of the Danish Residential Mobility dataset. The gender imbalance is the first panel due to the 1.07 male/female ratio rate at birth[55]. Indeed, we study the total population, not the currently alive population, that is balanced for gender due higher death rate for men. Therefore, although there is an imbalance in gender, the dataset is not biased. The second panel shows the birth year of every individual in the dataset, it closely follows the birth rate of Denmark, and the differences are due to migrations[56].

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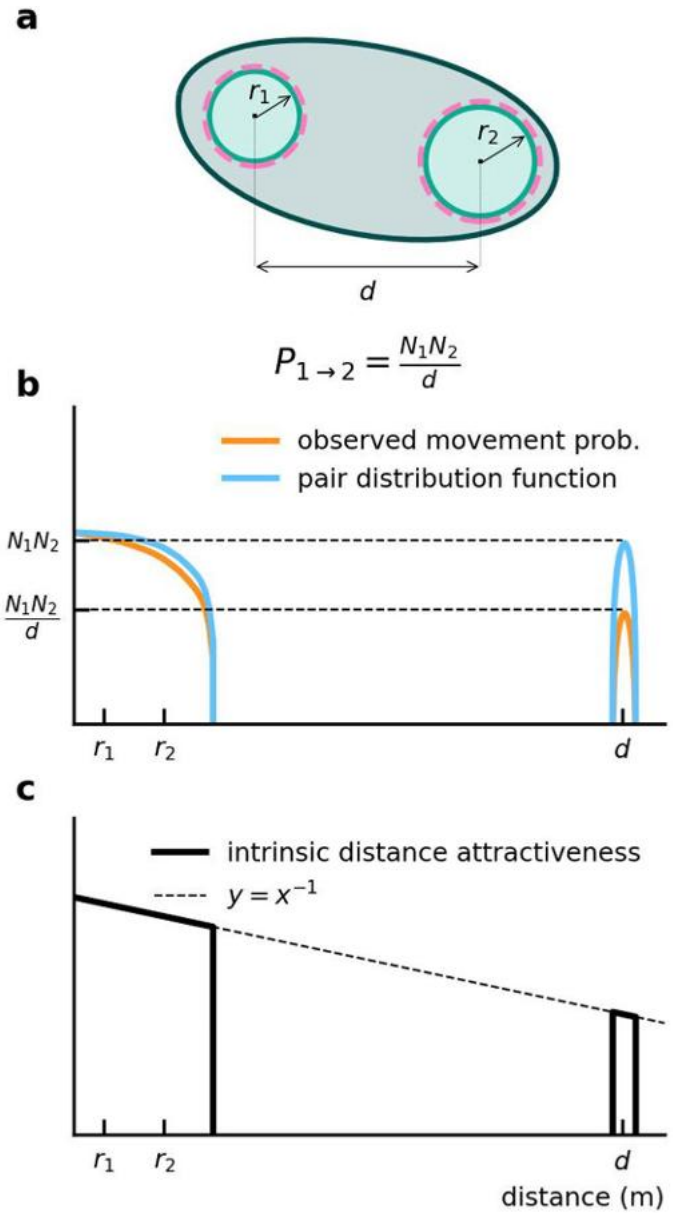
Extended Data Fig. 3 | Age Distribution. Distribution of Age at the time of moving in the Danish residential mobility data (see Methods: Data).

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Extended Data Fig. 4 | Comparison between our framework, the gravity model and the radiation model for cities.

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Extended Data Fig. 5 | A continuous gravity law.(**a**) This framework can be represented as a *continuous gravity law*, if we take a toy model of two distant cities, (**b**) the pair distribution at the distance *d* between the two cities collapses into a Dirac distribution, with the height equal to the product of cities population, N_1N_2. (**c**) The toy *intrinsic distance cost* of the simulation model. For a comparison with a continuous radiation model, see Supplementary Section 3.1.2.

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Extended Data Table 1 | Power law estimation

Type	Dataset	n	x_{min}	x_{max}	Exponent	KS	p
Residential Mobility	Denmark	39M	10	475032	**0.98**	0.01	**0.99**
	France	40M	3000	1134667	**1.07**	0.01	**0.98**
Day-To-Day Mobility	San Francisco	112168	10	55639	**0.94**	0.07	**0.96**
	Houston	47996	43	71209	**0.95**	0.03	**0.98**
	Singapore	15167	10	79674	**0.89**	0.12	**0.67**
Geography	City Radius	1367	2878	N/A	3.05	0.58	0.11

The table presents information about the estimation of the exponent of the power law. The estimation of the power law is based on a maximum-likelihood estimator as in[64]. The values of x_{min} and x_{max} are reported in meters. The table reports also the p-values associated with the Kolmogorov-Smirnov test for the power law exponent estimators[64]. According to[64], the suitability of a power law model is considered statistically implausible if the p-value is less than or equal to 0.1. This threshold indicates a less than or equal to 10% probability that the observed deviation of the data from the model predicions is due to random variation alone. Therefore the power-law model is statistically significant if $p > 0.1$ (in bold).

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Extended Data Table 2 | Power law log-likelihoods

Type	Dataset	Log-Normal	Exponential	Weibull
Residential Mobility	Denmark	128697	19209	89676
	France	5042	411	18535
Day-to-day Mobility	San Francisco	1091	91	2001
	Houston	446	35	2946
	Singapore	729	104	3651
Geography	Mobility City Radius	−0.96	1.91	1.18

The table presents the log-likelihood ratio $\mathcal{R}$[64] comparing the power law (cf. Extended Data Table 1) to other heavy-tailed distributions (one per column). When $\mathcal{R}$ is positive, the power law distribution has a higher likelihood compared with the alternative. When $\mathcal{R}$ is negative, the other distribution has a higher likelihood compared with the power law.

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Data collection: No software was used to collect the data

Data analysis	Software and Tools Used: All analyses were conducted using Python 3.11, with the following key packages and versions: matplotlib==3.4.3 shapely==1.7.1 pandas==1.3.0 scipy==1.7.0 powerlaw==1.5 fincoretails==0.0.1 Custom Code: Custom code for generating building locations using the models described in the main text (including simulations of 2D Lennard-Jones spheres in an external potential via the Velocity-Verlet algorithm, molecular dynamics, collision detection, and Berendsen thermostat) is available at: https://github.com/benmaier/ljhouses Code for estimating power-law behavior can be accessed at: https://github.com/lcb0b/role-of-geo Zenodo Archive: All custom code is also available on Zenodo: https://zenodo.org/records/14329837

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The data for Denmark used in this study are not publicly available due to Danish Data Protection regulations. However, access to the data is possible for research purposes. Data can be obtained via Statistics Denmark for Researchers in accordance with Statistics Denmark's Research Scheme: https://www.dst.dk/en/TilSalg/Forskningsservice. The location data for France are publicly available at https://adresse.data.gouv.fr/donnees-nationales and https://www.data.gouv.fr. The residential mobility data were obtained from https://www.insee.fr/fr/statistiques. The Foursquare data are publicly available at https://sites.google.com/site/yangdingqi/home/foursquare-dataset. The source data used to generate the figures are available via the Zenodo repository: https://zenodo.org/records/14329837.

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Study description	Overall the analysis is perform on existing datasets. The main dataset was collected by the national registry in Denmark. Two smaller datasets were collect by the national registry in France and Foursquare respectively. The study design is quantitative, utilizing large-scale, anonymized datasets from national registries in Denmark and France, along with location-based data from the Foursquare platform.
Research sample	Denmark. The dataset includes every residential move in Denmark from 1986 to 2020, making it fully representative of the population. The data come from high-quality national administrative registries, which record all address changes. France. The French dataset captures 40 million inter-city moves (highest resolution publicly available) from 2016 to 2020, sourced from INSEE. Location data were derived from the Base Adresse Nationale and housing statistics from FiLoSoFi, enabling detailed spatial mapping of residential addresses. Foursquare. Daily mobility data were drawn from publicly shared user check-ins on Foursquare (May–Nov 2010), accessed via Twitter's API. The dataset includes ~240,000 movements across three cities.
Sampling strategy	No sampling procedure was applied. The study relies on complete population-level records. The Foursquare dataset consists of data from Foursquare users.
Data collection	Data collection was based entirely on existing, administrative, or publicly available digital records.
Timing	Denmark. The dataset covers all residential moves recorded in Denmark over a 35-year period, from 1986 to 2020. France. The French residential mobility data cover inter-city migrations from 2016 to 2020. Foursquare. The daily mobility data were collected over a six-month window, from May 27, 2010 to November 3, 2010.
Data exclusions	In the case of the Danish residential mobility dataset, residential stays of less than one week were excluded from the analysis. This exclusion affected 4034 records, accounting for less than 0.1% of the total dataset.
Non-participation	N/A
Randomization	Randomization was not applicable to the study because it is based entirely on observational, retrospective data from administrative records and public datasets.

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